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| 17/167,404                                                                         | 02/04/2021  | Robert Ernest Troxler | 202-11-REI                   | 3345             |
| 79683                                                                              | 7590        | 01/09/2026            |                              |                  |
| OLIVE LAW GROUP, PLLC<br>BENTLEY J OLIVE<br>2500 Regency Parkway<br>CARY, NC 27518 |             |                       | EXAMINER<br>SANDERSON, LEE E |                  |
|                                                                                    |             |                       | ART UNIT                     | PAPER NUMBER     |
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**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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## Office Action Summary

**Application No.**

17/167,404

**Applicant(s)**

Troxler, Robert Ernest

**Examiner**

LEE E SANDERSON

**Art Unit**

3991

**AIA (First Inventor to File) Status No**

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1) ☒ Responsive to communication(s) filed on 10 November 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on \_\_\_\_.

2a) ☒ This action is **FINAL**.

2b) ☐ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on \_\_\_\_; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims\***

5) ☒ Claim(s) 1-41 is/are pending in the application.

5a) Of the above claim(s) \_\_\_\_ is/are withdrawn from consideration.

6) ☐ Claim(s) \_\_\_\_ is/are allowed.

7) ☒ Claim(s) 1-41 is/are rejected.

8) ☒ Claim(s) 1,3,5,7-12,15-17,20-21,24,32-33,35 and 37 is/are objected to.

9) ☐ Claim(s) \_\_\_\_ are subject to restriction and/or election requirement

\* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see [http://www.uspto.gov/patents/init\\_events/pph/index.jsp](http://www.uspto.gov/patents/init_events/pph/index.jsp) or send an inquiry to [PPHfeedback@uspto.gov](mailto:PPHfeedback@uspto.gov).

**Application Papers**

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 08 February 2021 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a) ☐ All      b) ☐ Some\*\*      c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. \_\_\_\_.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

\*\* See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date \_\_\_\_.

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: \_\_\_\_.

Paper No(s)/Mail Date \_\_\_\_.

## DETAILED ACTION

### *Final Rejection*

This is a reissue of application of U.S. Patent No. 10,197,686 ("the '686 Patent").

5 This application was filed 4 February 2021 and therefore the statutory provisions of the American Invents Act ("AIA") govern this proceeding. All references to 35 U.S.C. 251 and 37 CFR 1.172, 1.175, and 3.73 are to the current provisions. Also, in light of the effective filing date of the '686 Patent the pre-AIA first to invent provisions govern.

### 10 *Status of Claims*

As best as the Examiner can tell the status of the claims in the instant reissue application is as follows:

|    |                                                                 |                            |
|----|-----------------------------------------------------------------|----------------------------|
|    | <b>Claims 1 and 6:</b>                                          | (Original, thrice amended) |
|    | <b>Claim 3 :</b>                                                | (Original, twice amended)  |
| 15 | <b>Claims 4, 5, and 7:</b>                                      | (Original, amended)        |
|    | <b>Claim 2:</b>                                                 | (Original)                 |
|    | <b>Claims 8, 9, 11, 13, 15, 17, 21, 22, 24, 26, 29, and 32:</b> | (New, thrice amended)      |
|    | <b>Claims 14, 16, 20, 21, 22, 27, and 29:</b>                   | (New, twice amended)       |
|    | <b>Claims 12, 15, 18, 19, 23, 27, and 33-41:</b>                | (New, amended)             |
| 20 | <b>Claims 25, 26, 28, 31, 40, and 41:</b>                       | (New)                      |

### *Examiner Notes*

The Examiner notes that any objection and/or rejection previously set forth in the non-final Office action filed 8 May 2025 and not repeated herein is overcome and hereby withdrawn.

***Claim Objections***

**Claims 1, 3, 5, 7-12, 15-17, 20, 21, 24, 27, 32, 33, 35, and 37 are objected to because of the following informalities:**

**Claims 1-41** are objected for not complying with 37 C.F.R. 1.173(b)(2) due to improper status indicators. The proper status indicators for each of the claims is indicated above under the "Status of Claims" section. It is the Applicant's responsibility to correctly indicate the status of each of the claims (see MPEP 1453).

In **claims 1, 24, 32, and 33**, the words "Neon" and "Argon" should be written as "neon" and "argon" as the names of elements are common nouns, not proper nouns.

In **claim 1**, the word "whereas" recited in line 10 should be "wherein".

In **claim 1**, the phrase "adjust the waveform of the detector to form for measurement" is grammatically incorrect.

In **claims 3 and 35**, the term "penning gas" should be written as "Penning gas" as the name "Penning" is a proper noun.

In **claim 3**, the word "whereas" recited in line 4 should be "wherein".

In **claim 5**, the word "whereas" recited in line 4 should be "wherein".

In **claim 8**, the word "whereas" recited in line 8 should be "wherein".

In **claim 9**, the term " $^4\text{He}$ " recited in line 3 of claim 9 should be written as " $^4\text{He}$ " as this is the proper nuclear notation for isotopes.

In **claim 10**, the word "whereas" recited in line 3 should be "wherein".

In **claim 11**, the phrase "nanoparticles ,Boron nitride" in line 3 is grammatically incorrect and should be written as "nanoparticles, boron nitride" wherein there is a space after the comma and not before.

In **claim 12**, the word "whereas" recited in line 4 should be "wherein".

**Claim 15** recites "further coated in a neutron interacting material up to 1 mg/cm<sup>2</sup> of a Boron-10 or neutron capturing component" which is grammatically incorrect.

In **claim 15**, the word “whereas” recited in line 8 should be “wherein”.

In **claim 16**, the term “Halogens” should be written as “halogens” as the word “halogens” is a common noun.

5 In **claim 16**, the term “Halogen” should be written as “halogen” as the word “halogens” is a common noun

In **claim 16**, the term “Br” is repeated twice.

In **claim 20**, the word “carbide” should not be hyphenated.

In **claim 21**, in the second line there is an extraneous space between the term “ $^3\text{H}_3$ ” and the word “or”.

10 **Claims 21 and 32** recite the term “ $\text{He}_3$ ” which should be “ $^3\text{He}$ ” as  $\text{He}_3$  does not exist because helium is a noble gas.

In **claim 24**, the word “whereas” recited in line 8 should be the word “wherein”.

**Claim 27** is not properly marked and does not comply with 37 CFR 1.173. Claim 27 is a new claim, however the claim is not entirely underlined as required. It is noted that all claims  
15 must be presented relative to the patent and all things new to the patent must be underlined. See MPEP 1453 for guidance on how to make amendments in reissue. Improper amendments after final will result in non-entry and an advisory action.

In **claim 32**, the term “Penning” recited in line 5 should be “Penning gas”.

In **claim 32**, the word “whereas” recited in line 3 should be “wherein”.

20 In **claim 33**, the word “whereas” recited in line 8 should be “wherein”.

In **claim 35**, the word “whereas” recited in line 3 should be “wherein”.

In **claim 37**, the word “whereas” recited in line 4 should be “wherein”.

Appropriate correction is required.

***Claim Rejections - 35 USC § 112***

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. §112:

5           The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

10           The following is a quotation of the second paragraph of pre-AIA 35 U.S.C. §112:

          The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

15           **Claims 1-7, 11, 15-32, 40, and 41 are rejected under 35 U.S.C. § 112**

**(pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in**  
20 **the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. §112, the inventor(s), at the time the application was filed, had possession of the claimed invention.**

**The added material which is not supported by the prior patent is as follows:**

25           **Claim 1** recites “a predetermined mass thickness of 0.75-1.25 mg/cm<sup>2</sup> in Boron 10”. Similarly, claim 29 recites “coated with a range of 0.75mg/cm<sup>2</sup> to 1.25mg/cm<sup>2</sup> of substantially pure Boron-10”. However, there is no disclosure in the ‘686 Patent of a range of amount of 0.75 to 1.25 mg/cm<sup>2</sup>. Regarding the amount of <sup>10</sup>B in the interior of the gas tube, the ‘686 Patent merely discloses “Typical layers of Boron-10 which may  
30 be 1 mg/cm<sup>2</sup> but not limited to that dose” (col. 9 lines 1-2) which clearly does not disclose the claimed range of 0.75 to 1.25 mg/cm<sup>2</sup>.

It is noted that on page 11 of the remarks filed 10 November 2025, Applicant points to column 4, lines 25-40 of the '686 Patent for support for the amendments made to claim 1. However, the cited portion of the '686 Patent provides no disclosure regarding the amount of  $^{10}\text{B}$  in the interior of the tube of the detector. As such, there is

5 insufficient written description support for a mass thickness range of  $0.75\text{-}1.25\text{ mg/cm}^2$ .

**Claims 2-7, 40, and 41** are likewise rejected as depending from claim 1 and requiring all the limitations thereof.

**Claim 11** recites "at least one of a pure neutron interacting material including". However, there is no disclosure of a pure neutron interacting material in the '686 Patent.

10 As such, there is insufficient written description support for the claimed pure neutron interacting material.

**Claim 15** recites "up to  $1\text{ mg/cm}^2$  of a Boron 10 or neutron capturing component material". However, there is no disclosure of the claimed range of  $^{10}\text{B}$  or neutron capturing component material in the '686 Patent. Additionally, there is no disclosure in

15 the '686 Patent regarding a neutron capture component material. Furthermore, as it pertains to amounts of  $^{10}\text{B}$  or neutron capturing component material, the only amount recited in the '686 Patent is  $1\text{ mg/cm}^2$  of Boron-10 (col. 9 lines 1-2) and about  $1\text{ mg/cm}^2$  Boron-10 (claim 1). Furthermore, it is noted that on page 11 of the remarks filed 10 November 2025, Applicant points to column 4, lines 25-40 of the '686 Patent for support

20 for the amendments made to claim 15, the cited portion of the '686 Patent provides no disclosure regarding the amount of  $^{10}\text{B}$  or neutron capture component material in the interior of the tube of the detector. As such, there is insufficient written description support for a neutron capturing component material or for a range of up to  $1\text{ mg/cm}^2$ .

**Claims 16-23** are likewise rejected as depending from claim 1 and requiring all the limitations thereof.

**Claim 15** recites “3 to 93% amount of a neutron interacting gas”. However, there is no disclosure of 3 to 93% of a neutron interacting gas in the ‘686 Patent. As such, there is insufficient written description support for this limitation. If this value is derived from something that is disclosed in the ‘686 Patent, Applicant should explain how the claimed value is derived. **Claims 16-23** are likewise rejected as depending from claim 15 and requiring all the limitations thereof.

**Claim 17** recites “the component comprises at least one of a Halogen gas, CO<sub>2</sub> gas, active and passive RC or external circuit”. However, in the ‘686 Patent there is only written description support for a passive RC circuit. There is no disclosure in the ‘686 Patent of a quench component comprising an active and a passive RC or an external circuit.

**Claim 18** recites “coated with about 1 mg/cm<sup>3</sup>”. However, there is no disclosure of a coating measured in density units (i.e., mg/cm<sup>3</sup>) in the ‘686 Patent. As such, there is insufficient written description support for this limitation.

Regarding **claims 21 and 22**, there is no disclosure in the ‘686 Patent regarding the presence of <sup>3</sup>H<sub>3</sub>. As such, there is insufficient written description support for this limitation.

**Claim 22** recites “from less than 1 atmosphere to about 5 atmospheres of <sup>3</sup>H<sub>3</sub> or less than 1 atmosphere of BF<sub>3</sub>”. However, there is no disclosure in the ‘686 Patent regarding a pressure range of <sup>3</sup>H<sub>3</sub> being going as high as 5 atmospheres. As such, there is insufficient written description support for this limitation.



**Claim 24** recites “neutron capturing coated straws”. However, there is no disclosure of neutron capturing coated straws in the ‘686 Patent. It is noted that the ‘686 Patent does disclose a straw detector (claim 7) and straw tubes (col. 4 line 3) these recitations would not lead one of ordinary skill in the art to a coated straw which captures neutrons. As such, there is insufficient written description support for the claimed neutron capturing coated straw. **Claims 25-32** are likewise rejected as depending from claim 24 and requiring all the limitations thereof.

**Claims 3, 13, 15, 23-32, 38, and 40 are rejected under 35 U.S.C. § 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the Applicant regards as the invention.**

**Claim 3** has been amended so as to recite “coated in a substantially Boron-10” which renders the claim indefinite as it is unclear what this phrase means. Is the claimed at least one gas tube substantially coated in Boron-10? Or, is the claimed at least one gas tube coated with a substantially pure Boron-100? Upon review of the specification of the ‘686 Patent it appears that the claim is intended to recite “coated in a substantially pure Boron-10”. For the purpose of examination, the Examiner will interpret claim 3 as if it recites “at least one gas tube is coated in a substantially pure Boron-10”.

**Claims 13, 23, and 38** recite the term “the cathode.” There is insufficient antecedent basis for this limitation as no cathode is introduced in these claims or the claims from which they depend.

**Claim 15** recites “a neutron interacting material up to 1 mg/cm<sup>2</sup> of a Boron 10” which renders the claimed indefinite as it is unclear if this limitation is meant to mean

that the neutron interacting material contains up to 1 mg/cm<sup>2</sup> of a Boron 10 or if the neutron interacting material as a whole is present in material amount of up to 1 mg/cm<sup>2</sup>.

**Claim 15** recites the limitation “the ionization and quench components” for which there is insufficient antecedent basis for in the claim. While the quench component has antecedent basis, there is no such basis for an “ionization component” as it was referred to earlier in the claim as just “an ionizing gas.” These should be made consistent, either call the first instance “an ionization component” or call the latter instance “the ionizing gas and quench component.”

**Claim 15** recites the limitation “the at least one gas detector” in line 4. There is insufficient antecedent basis for this limitation in the claim as there is no prior recitation of a gas detector. As such, one of ordinary skill in the art would not be reasonably apprised of what claimed element corresponds to the recited gas detector.

**Claims 15 and 33** both recite the phrase “quench components” in the second to last line of each claim. Earlier in the claim these were referred to as “a quench component.” So, it is not clear if these components should be singular plural, this is an antecedent basis problem. These should be made consistent.

**Claims 21 and 22** refer to “<sup>3</sup>H<sub>3</sub>”. It is not clear what this means, so it is indefinite. This is likely a typographical error and should be changed to <sup>3</sup>He like parent claim 15.

**Claim 24** recites that the detector gas contains “a substantially reduced amount of a neutron absorbing [or in some claims interacting] gas.” This begs the question—substantially reduced from what, and what is the resulting amount that is the substantially reduced amount? When read in the context of the specification a person skilled in the art would understand that this means a substantially reduced amount from

the normal amounts found in typical detectors. The specification explains that the entire purpose of the patent is to reduce the amount of He used in such detectors as compared to the amount typically used. The specification also explains that while BF<sub>3</sub>-based detectors have been known as a substitute for He-based, BF<sub>3</sub> is toxic and it would likewise be useful to reduce the amounts used. So, when saying “substantially reduced” the claim is clearly referring to an amount reduced from the amounts used typically, which the specification further tells us is normally 70 cmHg for BF<sub>3</sub>, col. 8 lines 54-56, so to meet the claim the amount used must be substantially reduced from those amounts.

But what does that mean it should be reduced *to*? The specification only tells us that with a boron lined tube the pressure of BF<sub>3</sub> may be reduced to 5-70 cmHg. Col. 9 lines 27-30. This range, which *includes the normal value* of 70 cmHg, does not tell us what constitutes “substantial” reduction. Indeed it permits 70, or values very close to 70, so does not tell us much at all. Is 60 a substantial reduction? Must it get all the way down near 5 to be substantial? The problem is that the specification does not tell us, there is no way to determine the metes and bounds of the claim. This rejection would be overcome if “substantially” is removed from the independent claims. Merely saying “reduced” would be understood as being reduced from the typical value, as the specification explains. **Claims 25-32** are likewise rejected as depending from claim 24

**Claim 24** recites the limitations “detector gas” and “the ionization gas components” in the lines 4, 8, and 9. There is insufficient antecedent basis for these limitations in the claim. There is no previous mention of the presence of a detector gas and thus it is unclear what detector gas is being referred to in the claim. Additionally, it

is noted that while claim 24 introduces the term “an ionizing gas” is it unclear if the ionizing gas is the same as the claimed ionizing gas component. **Claims 25-32** are likewise rejected as depending from claim 24

**Claim 24** recites the limitations “detector gas” and “the ionization gas components” in line 5. There is insufficient antecedent basis for these limitations in the claim as there is previous mention of the presence of a detector gas or an ionizing components. As such, it is unclear what detector gas and ionizing gas components are being referred to in the claim. **Claims 25-32** are likewise rejected as depending from claim 24.

**Claims 27-29 and 31** recite the term “Boron gas tube” for which there is no antecedent basis. It is noted that the claimed “Boron gas tube” is not introduced in any of claims 27-29 and 31. Additionally, it is noted that independent claim 24 from which claims 27-29 and 31 depend does not recite “a Boron gas tube” rather it recites “a neutron interacting gas tube”. It is unclear if the Boron gas tube recited in claims 27-29 and 31 is the same as the neutron interacting gas tube recited in claim 24.

**Claim 32** in line 4 states “the non-interacting gas mix contains...” There is no antecedent basis for this term. Previously only “a gas mix” or “a mix which has no specific interaction...” were claimed. As such, it is unclear what is being referred to by the term “the non-interacting gas mix’.

**Claim 40** recites “the at least one detector is coated of about 1mg/cm<sup>2</sup> of the neutron absorbing gas tube”. This recitation is grammatically awkward and/or incorrect and appears to indicated that the at least one detector is coated with a neutron absorbing glass tube. Based on the understanding of one of ordinary skill in the art and

from the disclosure of the '686 Patent it is unclear how the at least one detector can be coated with a neutron absorbing glass tube. Additionally, it is unclear if the coated at least one detector recited in claim 40 is the same as the at least one gas tube recited in claim 1 or if the coated at least one detector is an additional element. As such, claim 40 is indefinite.

***Claim Rejections - 35 USC § 251***

The following is a quotation of the first paragraph of 35 U.S.C. 251:

(a) IN GENERAL - Whenever any patent is, through error, deemed wholly or partly inoperative or invalid, by reason of a defective specification or drawing, or by reason of the patentee claiming more or less than he had a right to claim in the patent, the Director shall, on the surrender of such patent and the payment of the fee required by law, reissue the patent for the invention disclosed in the original patent, and in accordance with a new and amended application, for the unexpired part of the term of the original patent. No new matter shall be introduced into the application for reissue.

**Claims 1-7, 11, 15-32, 40, and 41 are rejected under 35 U.S.C. § 251 because section 251 precludes new matter from being presented in a reissue application. New matter is present as describe in the rejections under 35 U.S.C. § 112 first paragraph present above.**

***Recapture***

**Claims 1-32, 40, and 41 are rejected under 35 U.S.C. § 251 as being an improper recapture of broadened claimed subject matter surrendered during the prosecution of the original application upon which the present reissue is based.**

See Greenliant Systems, Inc. et al v. Xicor LLC, 692 F.3d 1261, 103 USPQ2d 1951 (Fed. Cir. 2012); In re Shahram Mostafazadeh and Joseph O. Smith, 643 F.3d 1353, 98 USPQ2d 1639 (Fed. Cir. 2011); North American Container, Inc. v. Plastipak Packaging, Inc., 415 F.3d 1335, 75 USPQ2d 1545 (Fed. Cir. 2005); Pannu v. Storz Instruments

Inc., 258 F.3d 1366, 59 USPQ2d 1597 (Fed. Cir. 2001); Hester Industries, Inc. v. Stein, Inc., 142 F.3d 1472, 46 USPQ2d 1641 (Fed. Cir. 1998); In re Clement, 131 F.3d 1464, 45 USPQ2d 1161 (Fed. Cir. 1997); Ball Corp. v. United States, 729 F.2d 1429, 1436, 221 USPQ 289, 295 (Fed.Cir.). 1984); MBO Laboratories, Inc. v. Becton, Dickinson, & Co., 602 F.3d 1306, 1316-1317, 94 USPQ2d 1598 (Fed. Cir. 2010).

The "original application" includes the patent family's entire prosecution history. MBO Laboratories, Inc. v. Becton, Dickinson & Co., 602 F.3d 1306, 94 USPQ2d 1598 (Fed. Cir. 2010). A broadening aspect is present in the reissue which was not present in the application for patent. The record of the original application for the patent shows that the broadening aspect (in the reissue) relates to claimed subject matter that Applicant previously surrendered during the prosecution of the application. Accordingly, the narrow scope of the claims in the patent was not an error within the meaning of 35 U.S.C. 251, and the broader scope of claimed subject matter surrendered in the original application for the patent cannot be recaptured by the filing of the present reissue application.

In light of the case law, the MPEP sets forth a three step test for the recapture analysis which laid out below (see MPEP 1412.02 II):

**Step 1: Whether the reissue claims are broader in scope than the original patent claims.**

Instantly pending independent **claim 1** recites "coated with a predetermined mass thickness of 0.75-1.25 mg/cm<sup>2</sup> in Boron 10" which is broader in scope than the "coated optimally of about 1 mg/cm<sup>2</sup> in Boron 10" recited in claim 1 of the '686 Patent. **Claim 1** additionally recites "including partial pressures of at least one of <sup>3</sup>He and BF<sub>3</sub>"

which is broader in scope than the recitation of “including partial pressures of at least  $^3\text{He}$  and  $\text{BF}_3$ ” in claim 1 of the ‘686 Patent.

Instantly pending **claim 8** recites “including partial pressures of at least one of  $^3\text{He}$  and  $\text{BF}_3$ ” which is broader in scope than the recitation of “including partial pressures of at least  $^3\text{He}$  and  $\text{BF}_3$ ” in claim 1 of the ‘686 Patent.

Instantly pending **claim 15** recites “up to  $1 \text{ mg/cm}^2$  of a Boron 10 or a neutron capturing component material” which is broader in scope than the “coated optimally of about  $1 \text{ mg/cm}^2$  in Boron 10” recited in claim 1 of the ‘686 Patent. **Claim 15** additionally recites including partial pressures of at least one of  $^3\text{He}$  and  $\text{BF}_3$ ” which is broader in scope than the recitation of “including partial pressures of at least  $^3\text{He}$  and  $\text{BF}_3$ ” in claim 1 of the ‘686 Patent.

Instantly pending **claim 24** does not recite a tube comprising a coating of Boron 10. As such, **claim 24** is inherently broader in scope than claim 1 of the ‘686 Patent in this aspect. **Claim 24** additionally recites including partial pressures of at least one of  $^3\text{He}$  and  $\text{BF}_3$ ” which is broader in scope than the recitation of “including partial pressures of at least  $^3\text{He}$  and  $\text{BF}_3$ ” in claim 1 of the ‘686 Patent.

**Step 2: Whether the broader aspects of the reissue claimed relate to subject matter surrendered in the original prosecution.**

During the prosecution of the application 15/823,266 which matured into the ‘686 Patent, claim 1 was amended so as to recite “optimally of about  $1 \text{ mg/cm}^2$ ”, “including partial pressures of at least  $^3\text{He}$  and  $\text{BF}_3$  and a quench component”, and “whereas the

argon, neon, and quench components increase the total pressure and improve the waveform of the detector output” in response to a prior art rejection set forth in non-final Office. The amendments to claim 1 led to the issuance of the patent. As such, these limitations relate to surrendered subject matter and each constitutes a surrender  
5 generating limitation (here after “SGL”).

**Step 3: Whether the reissue claims were materially narrowed in other aspects, so that claims may not have been enlarged, and hence avoid the recapture rule (see MPEP 1412.I.C.)**

10 **Regarding instantly pending claim 1**, it is noted the claim recites the limitation of “thickness of 0.75-1.25 mg/cm<sup>2</sup> in Boron 10” which is a narrowing amendment relative to the surrendered subject matter. **Claim 1** further recites “a reduced amount of a neutron absorbing gas component including partial pressures of at least one of <sup>3</sup>He and BF<sup>3</sup>” which is another narrowing amendment relative to the surrendered subject  
15 matter. Thus, the SGL has not been entirely eliminated from the claim. As such, the it must be determined if the retained portion of the SGL materially narrow the original claims to avoid capture (see MPEP 1412 (B)(2)).

As to the <sup>10</sup>B coating amount recited in instant claim 1, the prior art establishes the claimed thickness was well known. For example, US 2005/0258373 to Lacy  
20 discloses neutron detector which includes a gas chamber comprising an interior <sup>10</sup>B coating layer in an amount of from about 0.12 mg/cm<sup>2</sup> to about 1.2 mg/cm<sup>2</sup> [abstract, 0014, 0043] which encompasses the claimed range. Lacy further discloses an optimal coating mass thickness of 1 mg/cm<sup>2</sup> [0039] which lies within the claimed range. Furthermore, US 2003/0213917 to Menlove et al. disclose a neutron detector which



includes a gas filled chamber having an interior  $^{10}\text{B}$  coating layer having a mass thickness of from 0.01 to 1.0 mg/cm<sup>2</sup> [abstract, 0015-0017] which encompasses the mass thickness recited in claim 1.

As to the claimed neutron absorbing gas being at least one of  $^3\text{He}$  and  $\text{BF}_3$ , these  
5 neutron absorbing gases were well known in the prior art for use in neutron detectors. For example, US 2011/0114848 to Frank et al. discloses a neutron-sensitive detector which includes a gas chamber comprising a mixture of gases which includes  $^3\text{He}$  gas or  $\text{BF}_3$  gas [abstract, 0017, 0020, 0027].

In light of the cited prior art, it is evident that the retained portion of the SGL does  
10 not materially narrow instantly pending claim 1 and thus improper recapture has not been avoided (see MPEP 1412 (B)(2)). **Claims 2-7, 40, and 41** are likewise rejected as depending from claim 1.

**Regarding instantly pending claim 8**, it is noted that claim recites “a neutron  
15 interacting gas component including partial pressures of least one of  $^3\text{He}$  and  $\text{BF}_3$ . As such, claim 8 retains a portion of the SGL. However, as is described above for claim 1, 2011/0114848 to Frank et al. establishes that the use of  $^3\text{He}$  and  $\text{BF}_3$  as a neutron interacting gas in gas mixtures used in neutron detection device was well known in the prior art.

Thus it is evident that the retained portion of the SGL does not materially narrow  
20 instantly pending claim 8 and thus improper recapture has not been avoided (see MPEP 1412 (B)(2)). **Claims 9-14** are likewise rejected as depending from claim 8.

**Regarding instantly pending claim 15**, it is noted the claim recites “at least one gas detector is further coated in a neutron interacting material up to 1 mg/cm<sup>2</sup> of a Boron 10” which is a narrowing amendment relative to surrendered subject matter.

**Claim 15** further recites “a neutron interacting gas component comprising partial pressures of at least one of <sup>3</sup>He and BF<sub>3</sub>” which another narrowing amendment relative to surrendered subject matter. Thus, the SGL has not been entirely eliminated from the claim. As such, the it must be determined if the retained portion of the SGL materially narrow the original claims to avoid capture.

As to the <sup>10</sup>B coating amount, as is described above when addressing claim 1, US 2005/0258373 to Lacy and US 2003/0213917 to Menlove et al. serve as evidence that the claimed coating amount was well known in the art. As to the claimed neutron interacting gas component being at least one of <sup>3</sup>He and BF<sub>3</sub>, US 2011/0114848 to Frank et al. serves as evidence that the use of <sup>3</sup>He gas and BF<sub>3</sub> gas as neutron interacting gases in neutron detector was well known in the art. Thus, in light of the cited prior art, it is evident that the retained portion of the SGL does not materially narrow instantly pending claim 15 and thus improper recapture has not been avoided (see MPEP 1412 (B)(2)). **Claims 16-23** are likewise rejected as depending from claim 15.

**Regarding instantly pending claim 24**, it is noted that the claim does not include any limitations pertaining to a coating of Boron-10 being present in the interior of the tube in a specific amount. As such, there is no narrowing amendments relative to the surrendered subject matter. Rather here, by reissue, Applicant seeks to recapture substantially the entire subject matter (i.e., a tube which does not comprise an interior coating of Boron-10) that was surrendered in the prosecution of the original application.

Accordingly, the recapture rule cannot be avoided since independent claim 24 of the instant reissue application is not materially narrowed in any other aspects relative to the surrendered subject matter.

5 Additionally, it is noted that claim 24 recites “a neutron interacting gas component comprising partial pressures of at least one of  $^3\text{He}$  and  $\text{BF}_3$ ” which is another narrowing amendment relative to surrendered subject matter. Thus, this SGL has not been entirely eliminated from the claim. However, US 2011/0114848 to Frank et al. serves as evidence that the use of a  $^3\text{He}$  gas or  $\text{BF}_3$  gas as a neutron interacting gases in neutron detector was well known in the art. Thus, in light of the cited prior art, it is evident that  
10 the retained portion of this SGL does not materially narrow instantly pending claim 15 and thus improper recapture has not been avoided (see MPEP 1412 (B)(2)). **Claims 25-32** are likewise rejected as depending from claim 24.

### ***Claim Rejections - 35 USC § 103***

15 The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

20 (a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

**Claims 1-3, 6-10, 13-19, 21-35, and 38-40 are rejected under pre-AIA 35**  
25 **U.S.C. §103(a) as being unpatentable over US 2011/0114848 to Frank et al. (“Frank”) in view of US 2014/0117246 to Zhou et al. (“Zhou”), and further in view of US 2005/0258373 to Lacy (“Lacy”)(all references previously cited).**

Independent claims:

**Regarding claim 1:**

**1. A neutron detector comprising: at least one detector configured for detecting ionizing radiation wherein the at least one detector is coated in an active region interior of a neutron absorbing gas tube;**

Frank discloses a detector for detecting ionizing radiation wherein the detector may have neutron sensitivity (i.e., it is a neutron detector) [abstract, 0017, 0027, Fig. 1]. Fig. 1 of Frank shows a neutron absorbing gas tube [0017, Fig. 1]. The interior of the neutron absorbing gas tube includes a cathode material [0017, 0031]. Frank teaches that the inner surface of the tube may be substantially coated to achieve a desired property, and may include boron for neutron sensitivity [0017]. This would be an active region interior of the tube.

**the interior of the at least one gas tube is coated with a predetermined mass thickness of 0.75-1.25 mg/cm<sup>2</sup> of Boron 10;**

Frank teaches that the inner surface of the tube may be substantially coated to achieve a desired property, and may include boron for neutron sensitivity [0017].

Frank is silent regarding the boron being <sup>10</sup>B. Frank is also silent regarding the coating having a mass thickness of 0.75-1.25 mg/cm<sup>2</sup>.

Zhou discloses an apparatus for detecting neutron radiation (i.e., a neutron detector) wherein the detector comprises a gas chamber (102) wherein a cathode (106) and a fill gas are disposed within an interior volume (104) of the chamber [abstract, 0057, 0058, Fig. 1]. The cathode (106) comprises a conversion layer (110) disposed on the surface thereof wherein the conversion layer (110) may comprise <sup>10</sup>B [0060, Fig.

1]. Zhou teaches that the conversion layer (110) serves to convert neutrons into secondary charged particles which increases the efficiency of the detector [0060, 0062].

It would have been obvious to one of ordinary skill in the art at the time the instant invention was made to have coated the cathode material of the detector of Frank with

5 the  $^{10}\text{B}$ -containing conversion layer disclosed by Zhou with the expectation of increasing the efficiency of the detector.

Lacy discloses a neutron detector comprising gas filled tubes which are lined with a neutron conversion layer [abstract, 0013, 0014]. Lacy teaches that the neutron

conversion layer is based on a thin film of  $^{10}\text{B}$  and that an optimal thickness of the film is  
10  $1\text{ mg/cm}^2$  [0013, 0039.] In light of the teachings of Lacy, it would have been obvious to

one of ordinary skill in the art to have formed the  $^{10}\text{B}$ -containing conversion layer of the detector of modified Frank so as to have a mass thickness of  $1\text{ mg/cm}^2$  because this was known in the art to be an optimal thickness for conversion layers in neutron detectors.

15

**wherein the detector gas contains a mixture of Neon, Argon, a substantially reduced amount of a neutron absorbing gas component including partial pressures of at least one of  $^3\text{He}$  and  $\text{BF}_3$  and a quench component;**

Frank discloses that the detector gas may contain Ne, Ar, a quench component,  
20 and also may include  $^3\text{He}$  or  $\text{BF}_3$  [0020, 0027]. Because the  $^3\text{He}$  or  $\text{BF}_3$  would have been present as part of a mixture of gases, they would have inherently been present in a partial pressure.

**whereas the argon, neon and quench components increase the total pressure and adjust the waveform of the detector output.**

The references do not say that the Ar, Ne, and quench components increase the total pressure. However, adding more gases would increase the pressure and meet the claim. The references also do not say that the gases adjust the waveform of the detector output. However, the recitation of adjusting the waveform of detector output is merely claiming a property or result of the structure, and to the extent that the claimed structure is met, and here the prior art uses the same mixture of gases, one may presume that the claimed result is also inherent in the art and is met. See MPEP 2112.01 I.-II. (product claims having the same composition are presumed to have the same properties).

Regarding claim 8:

**8. A neutron detector comprising:**

**at least one detector configured for detecting ionizing radiation; wherein the at least one detector comprises a neutron absorbing gas tube; and**

**the interior of the at least one gas tube coated optimally with about 1 mg/cm<sup>2</sup> in Boron 10 or a neutron interacting component material;**

**wherein the detector further contains a mixture of at least one of a non-neutron interacting gas, Neon, Argon, a Penning gas, and/or a quench component, and a substantially reduced amount of a neutron interacting gas component including partial pressures of at least one of 3He and BF<sub>3</sub> and a quench component;**

**whereas the argon, neon and quench and non-neutron interacting gas components increase the total pressure and adjust the waveform of the detector output.**

**Claim 8** is broader than claim 1. All of the limitations it adds beyond what is in claim 1 are in “or” statements that are not necessarily required. Claim 8 is therefore rejected for the same reasons as claim 1.

Regarding claim 15:

**15. A neutron detector comprising:**

**at least one detector configured for detecting ionizing radiation; wherein the at least one detector includes a neutron interacting gas tube; and**

**the interior of the at least one gas tube is further coated in a neutron interacting material up to 1 mg/cm<sup>2</sup> of a Boron 10 or neutron capturing material;**

**wherein the detector gas contains partial pressures of an ionizing gas, and 3% to 93% amount of a neutron interacting gas component including partial pressures of at least one of <sup>3</sup>He and BF<sub>3</sub> and a quench component;**

**whereas the ionization and quench components increase the total pressure and adjust the waveform of the detector output.**

**Claim 15** is generally broader than claims 1 and 8. Claim 15 is therefore rejected for the same reasons. Furthermore, Zhou teaches that <sup>3</sup>He is expensive and that in order to reduce cost it is known to substitute a portion of <sup>3</sup>He with argon [0058]. Zhou suggests that 50% <sup>3</sup>He is a reasonable amount. [0058]. As such, it would have been obvious to use 50% as this reduces costs, as taught by Zhou. The 50% <sup>3</sup>He in the

gas of the resulting detector would have met the 3% to 93% limitation recited in claim 15.

Regarding claim 24:

- 5        **24. A neutron detector comprising:**
- at least one detector configured for detecting ionizing radiation; wherein**
- the at least one detector is coated in an active region interior of a neutron**
- interacting gas tube; and**
- the interior of the at least one gas tube contains at least one of neutron**
- 10      **capturing coated straws, coated cathodes, nanoparticles, nanopowders, and**
- Boron Nitride, increasing the active surface area of a neutron interacting material;**
- wherein the detector gas contains a mixture of an ionizing gas, and a**
- substantially reduced amount of a neutron interacting gas components including**
- partial pressures of at least one of  $^3\text{He}$  and  $\text{BF}_3$  up to 70 psi;**
- 15      **whereas the ionization gas components and pressures include at least one**
- of Argon, Neon, Penning, for a total pressure from less than 1 atmosphere up to**
- the total pressure.**

**Claim 24** is like claims 1 and 15, except that instead of the interior being coated with neutron interacting material the tube contains coated straws or a coated cathode or

20      nanoparticles or nanopowders of such material, which may not be in Frank. Lacy teaches that boron straw detectors are an alternative type of neutron detector [abstract]. It would have been obvious to one of ordinary skill in the art to use a straw tube as the gas tube because they can be made very small and thin, allowing them to be densely packed and providing high efficiency, as taught by Lacy [abstract, 0041]. These small,



thin, densely packed straws will “increase the surface area” of interacting material, as compared to a mere wall of a tube. As to “up to 70 psi,” this is shown in other claims below, keeping in mind that this has a lower bound of zero and therefore is very broad.

Ar, Ne, and Penning are shown as to claim 1. Note that last clause (“for a total

5 pressure...”) says nothing. It says that the total pressure may have a lower bound of less than 1 atm, which includes zero, and an upper bond of “the total pressure” which is basically a tautology—the gases have a total pressure up to the total pressure.

Regarding claim 33:

10 **33. A neutron detector comprising:**  
**at least one detector configured for detecting ionizing radiation;**  
**wherein the at least one detector is a neutron absorbing gas tube; and**

Frank discloses a detector for detecting ionizing radiation wherein the detector may have neutron sensitivity (i.e., it is a neutron detector) [abstract, 0017, 0027, Fig. 1].

15 Fig. 1 of Frank shows a neutron absorbing gas tube [0017, Fig. 1]. The interior of the neutron absorbing gas tube includes a cathode material [0017, 0031].

**the interior of the at least one gas tube is coated optimally of about 1**  
**mg/cm<sup>2</sup> in Boron 10;**

20 Frank teaches that the inner surface of the tube may be substantially coated to achieve a desired property, and may include boron for neutron sensitivity [0017]. Frank is silent regarding the boron being <sup>10</sup>B. Frank is also silent regarding the coating having a mass thickness of about 1 mg/cm<sup>2</sup>.

Zhou discloses an apparatus for detecting neutron radiation (i.e., a neutron detector) wherein the detector comprises a gas chamber (102) wherein a cathode (106) and a fill gas are disposed within an interior volume (104) [abstract, 0057, 0058, Fig. 1]. The cathode (106) comprises a conversion layer (110) disposed on the surface thereof wherein the conversion layer (110) may comprise  $^{10}\text{B}$  [0060, Fig. 1]. Zhou teaches that the conversion layer (110) serves to convert neutrons into secondary charged particles which increase the efficiency of the detector [0060, 0062]. It would have been obvious to one of ordinary skill in the art at the time the instant invention was made to have coated the cathode material of the detector of Frank with the  $^{10}\text{B}$ -containing conversion layer disclosed by Zhou with the expectation of increasing the efficiency of the detector.

Lacy discloses a neutron detector comprising gas filled tubes which are lined with a neutron conversion layer [abstract, 0013, 0014]. Lacy teaches that the neutron conversion layer is based on a thin film of  $^{10}\text{B}$  and that an optimal thickness of the film is  $1 \text{ mg/cm}^2$  [0013, 0039.] In light of the teachings of Lacy, it would have been obvious to one of ordinary skill in the art to have formed the  $^{10}\text{B}$ -containing conversion layer of the detector of modified Frank so as to have a mass thickness of  $1 \text{ mg/cm}^2$  because this was known in the art to be an optimal thickness for conversion layers in neutron detectors.

**wherein the detector gas contains a mixture of Neon, Argon, a reduced amount of a neutron absorbing gas component including partial pressure of at least  $^3\text{He}$  and  $\text{BF}_3$  and a quench component;**

Frank discloses that the detector gas may contain neon, argon, a quench component, [0020], and also may include  $^3\text{He}$  or  $\text{BF}_3$  for neutron sensitivity [0027].

Frank is silent regarding how much  $^3\text{He}$  or  $\text{BF}_3$  is used. Zhou suggests that 50%  $^3\text{He}$  is a reasonable amount. [0058]. It would have been obvious to use 50% as this reduces costs, as taught by Zhou. Additionally, while Frank is silent regarding utilizing a combination of  $^3\text{He}$  and  $\text{BF}_3$  in the detector gas, it is noted that MPEP 2144.06

5 establishes that it is *prima facie* obvious to combine two components each of which is taught by the prior art to be useful for the same purpose, in order to form a third composition to be used for the very same purpose. As such, given that Frank teaches that the detector gas may include  $^3\text{He}$  or  $\text{BF}_3$  for neutron sensitivity it would have been obvious to include both  $^3\text{He}$  and  $\text{BF}_3$ .

10 **whereas the argon, neon and quench components increase the total pressure and adjust the waveform of the detector output.**

The cited prior art references are silent regarding Ar, Ne, and quench components increasing the total pressure. However, adding more gases would

15 inherently increase the pressure and meet the claim. The cited references are also silent regarding the gases adjusting the waveform of the detector output. However, adjusting the waveform of detector output is merely claiming a property or result of the structure, and to the extent that the claimed structure is met, and here the prior art uses the same mixture of gases, one may presume that the claimed result is also inherent in

20 the art and is met. See MPEP 2112.01 I.-II. (product claims having the same composition are presumed to have the same properties).

Dependent claims:

**2. The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated in a substantially pure Boron-10 and also comprises a neutron absorbing gas and a Penning gas for adjusting the pressure and electronic waveform.**

5           **9. The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated in a substantially pure Boron-10 and also comprises a neutron interacting gas and at least one of Ar, Ne,  $^4\text{He}$ , Br,  $\text{CO}_2$ , or a Penning gas for adjusting the pressure and electronic waveform.**

As discussed above regarding claim 1 the tube is coated in Boron 10. Frank  
10 further discloses that its detector gas may  $\text{BF}_3$  or  $^3\text{He}$  (i.e. a neutron interacting gas), as well as Ar and/or Ne [0020, 0027]. Including different gases would inherently adjust any pressure and electronic waveform.

Frank is silent regarding the Boron-10 being “substantially pure.”

However, Lacy teaches that when a Boron 10 coating is used in detectors it may  
15 be employed in its natural form, which is generally 20% Boron 10, or it may be used in an enriched form up to 100%. [0015]. At 100%, this would be “substantially pure.” It would have been obvious to use 100% as it provides more sensitivity, allowing us to reduce the amount needed, as taught by Lacy.

20           **3. The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated in a substantially pure Boron-10 and also comprises a mix of a Penning gas and a  $\text{BF}_3$  gas whereas the  $\text{BF}_3$  gas is less than 700 Torr in pressure for absorbing thermal neutrons and the penning gas adjusts the waveform signal and gain.**

**10. The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated in an enriched Boron-10 and also comprises a mix of at least one of Ar, Ne, CO<sub>2</sub>, <sup>4</sup>He, Br, or Penning gas and a BF<sub>3</sub> gas whereas the BF<sub>3</sub> gas is less than 700 Torr in partial pressure for interacting with thermal neutrons and the fill gas adjusts the waveform signal and gain.**

As discussed above regarding claims 1 and 2, the tube is coated in Boron 10 and may be pure.

Lacy teaches that when a Boron 10 coating is used in detectors it may be used in an enriched form up to 100% [0015].

Frank further discloses that its detector gas may BF<sub>3</sub> or <sup>3</sup>He (i.e. a neutron interacting gas), as well as Ar and/or Ne [0020, 0027]. Various pressures are disclosed in Frank that are less than 700 Torr [0025, 0029]. Including different gases would inherently adjust any waveform signal and gain.

**6. The neutron detector of claim 1, wherein the cathode of the at least one gas tube contains an alloy of Boron with a non-corrosive metal and further comprises a mix of a Penning gas and a BF<sub>3</sub> gas of less than 700 Torr for adjusting the waveform signal and efficiency.**

**13. The neutron detector of claim 8, wherein the cathode of the at least one gas tube is one of an alloy or composite of Boron with a non-corrosive metal and further comprises a mix of at least one of Ar, Ne, and Penning gas and a neutron interacting gas for adjusting the waveform signal and efficiency.**

As discussed above regarding claim 1 the tube is coated in Boron 10.

Frank further discloses that its detector gas may  $\text{BF}_3$  or  $^3\text{He}$  (i.e. a neutron interacting gas), as well as Ar and/or Ne [0020, 0027]. Various pressures are disclosed in Frank that are less than 700 Torr [0025, 0029]. Including different gases would inherently adjust any waveform signal and efficiency.

5           Zhou further teaches that the cathode may be a combination of boron carbide and aluminum(i.e., an alloy or composite of boron) [0061]. It would have been obvious to use Zhou's boron in place of Frank's as a simple substitution of one known element for another to yield predictable results. MPEP 2143 I.B. Frank uses boron but does not specify what particular type, and this is described in Zhou. The boron has the same  
10       purpose in each reference, and it would have been obvious to use Zhou's in place of Frank's, and the result would have been predictable as again the purpose is the same in both references.

**7. The neutron detector of claim 1, wherein the at least one gas tube is a  
15       straw detector of Boron and further comprises a mix of a Penning gas and at least one of a  $^3\text{He}$  and  $\text{BF}_3$  gas of less than 700T for adjusting the waveform signal and efficiency.**

**14. The neutron detector of claim 8, wherein the at least one gas tube  
interior is at least one of a straw detector, lined tubes, or a cathode coating of  
20       Boron and further comprises a mix of at least one of Ar, Ne, and a Penning gas with at least one of a  $^3\text{He}$  and  $\text{BF}_3$  gas for adjusting the waveform signal and efficiency.**

As discussed above regarding claim 1 the tube is coated in Boron 10. Frank further discloses that its detector gas may  $\text{BF}_3$  or  $^3\text{He}$  (i.e. a neutron interacting gas), as

well as Ar and/or Ne [0020, 0027]. Various pressures are disclosed in Frank that are less than 700 Torr [0025, 0029]. Including different gases would inherently adjust any waveform signal and efficiency.

Frank is silent regarding the detector tube being a straw detector. However, Lacy teaches that boron straw detectors are an alternative type of neutron detector [abstract]. It would have been obvious to use a straw tube as the gas tube because they can be made very small and thin, allowing them to be densely packed and providing high efficiency, as taught by Lacy [abstract, 0041].

**16. The neutron detector of claim 15, wherein the ionizing gas comprises at least one of Ar, Ne, Br, Halogens, CO<sub>2</sub>, Br, and a Penning gas.**

Frank uses these gases, see the rejection of claim 1 above.

**17. The neutron detector of claim 15, wherein the quench component comprises at least one of a Halogen gas, CO<sub>2</sub> gas, active and a passive RC external circuit.**

Frank states the quench component can be halogen gas [0021].

**18. The neutron detector of claim 15, wherein the at least one gas tube is coated with about 1mg/cm<sup>3</sup> or lined in B10 or made up of neutron interacting straws.**

**19. The neutron detector of claim 15, wherein the at least one gas tube is coated with at least one of Boron-10, Crystalline Boron-10, Boron-10 Enriched Boron carbide, Boron-10 enriched metal, Boride ceramics/Metal Borides and Li-6.**

All shown in the rejections above, see claim 1 for the coatings or claim 7 for the neutron interacting straws.

**21. The neutron detector of claim 15, wherein the partial pressures sum to less than about 70 psi of  $^3\text{H}_3$  or  $\text{BF}_3$ .**

**22. The neutron detector of claim 15, wherein the partial pressures sum to 70 pounds per square inch (PSI) or from less than 1 atmosphere to about 5 atmospheres of  $^3\text{H}_3$  or  $\text{BF}_3$ .**

It is not clear why Applicant insists on mixing all of these units, but in any case Frank [0025-0029] and Zhou [0058] each give pressures falling within the claimed ranges. Claim 21 “less than about 70 psi” has a lower limit of zero and therefore is quite broad. Claim 22 is likewise quite broad as “less than 1 atmosphere” has a lower limit of zero, and 5 atmospheres is 3800 Torr. This range is broader than the other claimed limitation “sum to 70 PSI” so the broader range in this alternative statement governs.

**23. The neutron detector of claim 15, wherein the cathode is boron carbide/aluminum matrix.**

See rejection of claims 6 and 13 above.

**25. The neutron detector of claim 24, wherein the mixture is tuned for parameters of count rate, starting voltage, breakdown, output pulse voltage, and plateau shape.**

The ‘686 Patent does not actually explain how this tuning occurs, it merely states that the Penning gas adjusts these things. The references likewise teach the use of a



Penning gas (see [0055] of Frank). The Penning gas used will “tune” such parameters as evidenced by the ‘686 Patent.

**26. The neutron detector of claim 24, wherein the at least one detector is a neutron interacting Boron based gas tube.**

See rejection of claim 1 above. Interacting is broader than absorbing.

**27. The neutron detector of claim 24, wherein the interior of the at least one Boron gas tube is coated in a substantially pure Boron-10 coating.**

**28. The neutron detector of claim 24, wherein the interior of the at least one Boron gas tube is coated in a substantially pure Boron-10 coating and also comprises a Penning gas for adjusting output parameters.**

See rejection of claims 1-2 above.

**29. The neutron detector of claim 24, wherein the interior of the at least one Boron gas tube is coated with a range of 0.75mg/cm<sup>2</sup> to 1.25mg/cm<sup>2</sup> of substantially pure Boron-10, a range of least 5 cmHg-70cmHg partial pressure of BF<sub>3</sub> and 70 psi of 3He of an interacting gas, and less than 1 atmosphere of a non-neutron interacting fill gas and quench gas.**

For the mg/cm<sup>2</sup> range, see rejection of claim 1 above, and claim 2 for “substantially pure.” For the pressures, Zhou teaches that the pressures may be 100-15200 Torr in total, and that the interacting part may be 50% and replacements (such as non-interacting and quench) may be 50%. [0058]. 140 Torr is within the range and would be 140 mm/Hg, so 50% of each, 70 mm, would fall within the claimed ranges.

**30. The neutron detector of claim 24, wherein the coating is applied through a thick film processes including painting it on, or more controlled and cleaner processes, as used in thin film technology such as sputtering, CVD, Plasma enhanced CVD, evaporation, magnetron sputtering, glow discharge processes, and conformal coatings, and the like.**

This limitation is entirely a product-by-process claim, and therefore patentability is based on the device itself, not the process of making. MPEP 2113. While a specific claimed process might imply a specific structure that is required, the Examiner fails to see how these various different and alternative processes require some structure that is not already found in the base claim. Furthermore, Lacy uses plasma vapor deposition, [0045]. It would have been obvious to use this process, to the extent it is required in the claim, as it provides a highly uniform coating, as taught by Lacy.

**31. The neutron detector of claim 8, wherein the at least one Boron gas tube comprises a mix of at least one of helium, boron trifluoride, argon, neon, carbon dioxide, Penning gas, and a halogen gas.**

**32. The neutron detector of claim 31, wherein the detector contains a neutron interacting material coating, a gas mix, and a mix which has no specific interaction with a neutron, whereas the interacting material contains pure Boron 10 and the interacting gas mix contains one of a  $\text{BF}_3$  and  $\text{He}_3$ ; furthermore the non-interacting gas mix contains at least one of  $\text{CO}_2$ , Argon, Neon, Br, Penning, an ionizing gas and a quench mechanism.**

See rejection of claim 1, showing the references having these various gases in the tube along with Boron 10, and claim 2 for the Boron 10 being pure.

**34. The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated in a substantially pure Boron-10 and also comprises a neutron absorbing gas and a Penning gas for adjusting the pressure and electronic waveform.**

As discussed above regarding claim 33 the tube is coated in Boron 10. Frank discloses that the detector gas may contain Ne, Ar, a quench component, and also may include  $^3\text{He}$  or  $\text{BF}_3$  [0020, 0027]. Including different gases would inherently adjust any pressure and waveform. It is not explicit that the Boron-10 is “substantially pure.”

however Lacy teaches that when a Boron 10 coating is used in detectors it may be employed in its natural form, which is generally 20% Boron 10, or it may be used enriched form up to 100% [0015]. At 100%, this would be “substantially pure.” It would have been obvious to use 100% as it provides more sensitivity, allowing us to reduce the amount needed, as taught by Lacy.

**35. The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated in a substantially Boron-10 and also comprises a mix of a Penning gas and a  $\text{BF}_3$  gas whereas the  $\text{BF}_3$  gas is less than 700 Torr in pressure for absorbing thermal neutrons and the penning gas adjusts the waveform signal and gain.**

As discussed above regarding claim 33 Frank discloses that the detector gas may contain Ne, Ar, a quench component, and also may include  $^3\text{He}$  or  $\text{BF}_3$  [0020, 0027]. Various pressures are disclosed in Frank that are less than 700 Torr. [0025], [0029]. Including different gases would inherently adjust any waveform signal and gain.

**38. The neutron detector of claim 33, wherein the cathode of the at least one gas tube contains an alloy of Boron with a non-corrosive metal and further comprises a mix of a Penning gas and a BF<sub>3</sub> gas of less than 700 Torr for**  
5 **adjusting the waveform signal and efficiency.**

As discussed above regarding claim 33 the tube is coated in Boron 10. Additionally, Frank discloses that the detector gas may contain Ne, Ar, a quench component, and also may include <sup>3</sup>He or BF<sub>3</sub> [0020, 0027]. Various pressures are disclosed in Frank that are less than 700 Torr [0025, 0029]. Including different gases  
10 would change the operation of the detector and thus adjust any waveform signal and efficiency. Zhou further teaches that the cathode may be a combination of boron carbide and aluminum. [0061]. It would have been obvious to use Zhou's boron in place of Frank's as a simple substitution of one known element for another to yield predictable results. MPEP 2143 I.B. Frank uses boron but does not specify what particular type,  
15 and this is described in Zhou. The boron has the same purpose in each reference, and it would have been obvious to use Zhou's in place of Frank's, and the result would have been predictable as again the purpose is the same in both references.

**39. The neutron detector of claim 33, wherein the at least one gas tube is a**  
20 **straw detector of Boron and further comprises a mix of a Penning gas and at least one of a <sup>3</sup>He and BF<sub>3</sub> gas of less than 700T for adjusting the waveform signal and efficiency.**

As discussed above regarding claim 33 the tube is coated in Boron 10. Additionally, Frank discloses that the detector gas may contain Ne, Ar, a quench

component, and also may include  $^3\text{He}$  or  $\text{BF}_3$  [0020, 0027]. Various pressures are disclosed in Frank that are less than 700 Torr [0025, 0029]. Including different gases would change the operation of the detector and thus adjust any waveform. Frank does not disclose the tube is a straw detector. Lacy teaches that boron straw detectors are an alternative type of neutron detector [abstract]. It would have been obvious to use a straw tube as the gas tube because they can be made very small and thin, allowing them to be densely packed and providing high efficiency, as taught by Lacy [abstract, 0041].

**40. The neutron detector of claim 1, wherein the at least one detector is coated of about  $1\text{mg}/\text{cm}^2$  of the neutron absorbing gas tube.**

As is described above for claim 1, Frank as modified by Zhou and Lacy teaches a mass thickness of the  $^{10}\text{B}$  coating of  $1\text{ mg}/\text{cm}^2$ .

**Claims 4, 5, 11, 12, 36, and 37 are rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Frank, Zhou, and Lacy as applied to parent claims 1 and 8, and 33, and further in view of US 2011/0284755 to Stradins et al. (“Stradins”)(previously cited).**

**4. The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a Penning gas and a  $\text{BF}_3$  gas less than 700 Torr for adjusting the waveform signal and efficiency.**

**5. The neutron detector of claim 1, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a**

**Penning gas and at least one of a reduced amount of  $^3\text{He}$  and a  $\text{BF}_3$  gas for adjusting the waveform signal and efficiency; whereas the pressure is reduced below 700 Torr.**

5       **11. The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated with at least one of a substantially pure neutron interacting material including Boron-10 nanoparticles, nano powders, spherical nanoparticles, nanoparticles, Boron nitride, Crystalline 10Boron, 10B Enriched Boron carbide, 10B enriched metal, Boride ceramics/Metal Borides, and 10B enriched Boron Trifluoride, and conformal coatings, and further comprises a mix**  
10       **of a non-interacting fill and quench gas and a neutron interaction gas for adjusting the waveform signal and efficiency.**

15       **12. The neutron detector of claim 8, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of at least one of Ar, Ne, a Penning gas and at least one of a reduced amount of  $^3\text{He}$  and a  $\text{BF}_3$  gas for adjusting the waveform signal and efficiency; whereas the  $\text{BF}_3$  pressure is reduced below 700 Torr.**

20       **36. The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a Penning gas and a  $\text{BF}_3$  gas less than 700 Torr for adjusting the waveform signal and efficiency.**

**37. The neutron detector of claim 33, wherein the interior of the at least one gas tube is coated with Boron-10 nanoparticles and also comprises a mix of a Penning gas and at least one of a reduced amount of  $^3\text{He}$  and a  $\text{BF}_3$  gas for**

**adjusting the waveform signal and efficiency; whereas the pressure is reduced below 700 Torr.**

As discussed above regarding claim 1 the tube is coated in  $^{10}\text{B}$ . As in claim 2, it may be “pure.” Frank further discloses that its detector gas may  $\text{BF}_3$  or  $^3\text{He}$  (i.e. a neutron interacting gas), as well as a Penning gas, Ar and/or Ne [0020, 0027]. Various pressures are disclosed in Frank that are less than 700 Torr [0025, 0029]. Including different gases would change the operation of the detector and thus adjust any waveform. It is not disclosed that the boron-10 is “nanoparticles.” Stradins teaches that neutron detectors using a boron-10 film may use boron-10 nanoparticles to apply the boron [0009]. It would have been obvious to use nanoparticles as a simple substitution of one known element for another to yield predictable results. It would have been obvious to use nanoparticles and the result of the substitution would have been predictable because in either case the resulting device operates in substantially the same way, with the only potential difference being the way the boron-10 coating is formed.

**Claim 20 is rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Frank, Zhou, and Lacy as applied to parent claim 15, and further in view of U.S. 8,384,004 to Rowland et al. (“Rowland”)(previously cited).**

**20. The neutron detector of claim 15, wherein the Boron-10 is enriched Boron Trifluoride, Crystalline  $^{10}\text{B}$  Enriched Boron carbide,  $^{10}\text{B}$  enriched metal, Boride ceramics/Metal Borides.**

Boron-10 is used as in the rejection of the independent claims above, but it is not disclosed that it is enriched  $\text{BF}_3$ . Rowland teaches that in neutron detectors boron-10

may be used as enriched BF<sub>3</sub>. (col. 3, lines 1-24). It would have been obvious to do so as this provides for a higher amount of boron-10 as compared to natural boron, as taught by Rowland.

5           **Claim 41 is rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Frank, Zhou, and Lacy as applied to parent claim 1, and further in view of U.S. 2013/0067741 to Stephan et al. ("Stephan")(previously cited).**

**41. The neutron detector of claim 1, further comprising one or more temperature sensors configured to acquire temperature data for correction**  
10 **processes.**

          The references applied against claim 1 do not show a temperature sensor. Stephan teaches that a neutron detector may also include other sensors, including a temperature sensor [0110]. It would have been obvious to include a temperature sensor as Stephan recognizes it is useful to include to complement the neutron  
15 detecting capability of the device.

### ***Response to Arguments***

          The arguments presented in the Response filed 10 November 2025 have been fully considered but are not found persuasive.

20

### **Claim Objections**

          On page 10 of the Response Applicant asserts that the claims 1, 6, 8-41 are correctly marked. The Examiner disagrees with this assertion because as is described in the "Status of Claims" and "Claim Objection" sections above the status indicators of



the instantly pending claims do not reflect the actual amendment status of the claims.

Appropriate correction is required.

On page 10 of the Response Applicant asserts that the objection to claims 1, 6, and 33 for reciting terms “3HE” and/or “BF3” has been fully addressed. The Applicant further asserts that new claims 8-32 address the aforementioned issues. In response, it noted that as is described in the objections to the claims section of the instant Office action, the terms “3HE” and “BF3” are still recited variously throughout the claims and needs to be addressed.

10

#### **Rejections under 35 U.S.C. §251**

On page 10 of the Response Applicant asserts that claims 15-28 and 30 are amended such that they do not recapture broadened subject matter. This assertion is interpreted as meaning that claims 15-28 and 30 have been amended such that they do not improperly recapture surrendered subject matter.

15

In response to Applicant’s assertion, it is noted that claim 15 has been amended so as to recite “is coated in a neutron interacting material up to 1 mg/cm<sup>2</sup> of a Boron 10” and thus the surrendered generating limitation “coated optimally of about 1 mg/cm<sup>2</sup> in Boron 10” has been modified but not entirely eliminated. In such situations, a proper recapture analysis requires determination of whether or not the retained portion of the modified limitation is well known in the prior art. If the retained portion of the modified limitation is well known in the prior art, then impermissible recapture exists (see MPEP 1214.02 II.C.). In the instant case, coating amounts of <sup>10</sup>B in neutron detectors which fall within the claimed range were well known in the art and thus impermissible

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recapture still exists in claim 15 as amended. For the same or similar reasons, impermissible recapture also exists in claims 1, 8, and 24.

**Rejections under 35 U.S.C. §112, First Paragraph and 35 U.S.C §251 for new**

**5 matter.**

On pages 11 and 12 of the Response Applicant asserts that claims 15-30, 33-39, and 41 have been amended so as to comply with the written description requirement. In support this assertion, Applicant points to the table provided in the Response. Regarding claims 16, 17, 19-21, 23, 25-28, 33-39 and 41, it is noted that the no new  
10 matter has been indicated as being present in the instant Office action. However, the amendments made to claims 15, 18, 22, 24, and 29 incorporate subject matter for which there is insufficient written description support. The basis for this conclusion is described above.

Additionally, regarding independent claims 15 and 24, while Applicant provided  
15 table indicates that support for the amendments can be found at column 4, lines 25-40 of the '686 Patent, which has been reproduced below.

25 couple to an He3 gas tube 102. Further, the ionizing radiation detector circuit 100 is configured with at least one GM tube 110 coupled to a corresponding at least one port 112(0:N). The ionizing radiation detector circuit 100 operates biasing the He3 gas tube 102 to 900V as a non-limiting  
30 example. By biasing the He3 gas tube 102 to 900V a slow neutron interacting with the <sup>3</sup>He gas produces a negative going pulse in detection circuitry 114. In this example, the detection shaping circuitry 116 can invert and square up the produced waveform for an ADC counting circuit (not shown), or prepare for a peak amplitude histogram collection of events. Once the ionization event takes place, the He3 gas in the tube needs time to recombine during a "quenching process". This can take place with an RC circuit (not shown) or by selection of the correct species of gas in the tube.  
35  
40 With regard to FIG. 2, a block diagram showing a circuit

20

As can be seen, the cited section of the '686 Patent is directed towards the circuitry of the disclosed detector. The cited section of the '686 Patent does not describe any of the following limitation (A)-(C) recited in amended independent claim 15:

- 5 (A) "up to 1 mg/cm<sup>2</sup> of a Boron 10 or neutron capturing component material"
- (B) "the interior of the at least one gas detector is further coated in a neutron  
interacting material up to 1 mg/cm<sup>2</sup> of a Boron-10 or neutron capture  
component material"
- 10 (C) "3 to 93% amount of a neutron interacting gas"

In fact, sufficient written support for any of newly added limitations (A), (B), or (C) is not found anywhere in the '686 Patent.

- 15 Independent claim 24 recites the limitation of "neutron capturing coated straws" support for which is not found cited section (col. 4 lines 25-40) of the '686 Patent. A review of the '686 Patent reveals that sufficient written support for neutron capturing coated straws anywhere in the '686 Patent.

- Regarding claims 11, 18, 22, and 29 the sections of the '686 Patent cited by the  
20 Applicant in the Response as providing support for one or more of the newly added limitation do not provide sufficient written support and no such support is found anywhere in the '686 Patent. If Applicant feels that one or more of the rejection under 35 U.S.C. §112, first paragraph and 35 U.S.C. §251 for the introduction of new matter is improper, Applicant is encouraged to cite where specifically in the '686 Patent written  
25 description support for the amendment can be found and describe how the citation provides support. Additionally, if the asserted written description support is mathematically derived, Applicant is encouraged to explain the relevant derivation.

**Rejections under 35 U.S.C. §112, second paragraph**

On page 12 of the Response the Applicant asserts that claims 1 and 6 as amended and new claims 8-41 particularly and distinctly claim the subject matter regarded as the invention. It is agreed that the amendment made to claim 1 and new claims 8, 24, and 33 overcomes the rejection under 35 U.S.C. §112, second paragraph set forth in the prior Office action. However, claims 13, 26, 23-32, and 38 remain indefinite or have been rendered infinite by amendment for the reasons set forth above.

**Rejections under 35 U.S.C. §103**

On pages 13 and 16 of the Response Applicant asserts that neither Frank nor Zhou teaches or suggests the limitations recited in amended claim 1. However, neither Frank nor Zhou alone was relied upon to teach all the features recited in claim 1. Rather, Frank taken in combination with the teachings of Zhou and Lacy were and are relied upon to address all the features of claim 1. Regarding the Applicant's argument, it is noted that one cannot show nonobviousness by attacking references individually where the rejections are based on combinations of references (see MPEP 2145 IV). As such, Applicant's argument is not found persuasive.

On page 13 of the Response Applicant asserts that Frank does not really apply because it is only off handedly describing neutron tubes. But even off handed disclosure is disclosure and may be used in a rejection. Applicant does not show why Frank's disclosure cannot be applied against the claims. Additionally, MPEP 2123 I establishes that a reference may be relied upon for all that it would have reasonably suggested to one having ordinary skill in the art, including nonpreferred embodiments. In the instant

case, Frank explicitly describes neutron sensitive radiation detector comprising a gas tube having an inner surface which includes boron [0017] and comprising  $\text{BF}_3$  or  $^3\text{H}$  as a neutron sensitive gas component [0027]. As such, Frank clearly applies to the instantly claimed invention.

5           On pages 13-16 of the Response Applicant asserts that since Frank teaches that addition of air, moisture, and oxygen may be included in the disclosed detector it teaches away from neutron detectors. Applicant goes on to assert that neutron detectors can have no electronegative gases. In response, it is noted that Frank does not teach or even suggest that the disclosed neutron detector is required to comprise  
10   air, moisture, or oxygen. The presence of these gases is merely one embodiment of the ionizing radiation detection system disclosed by Frank. For example Frank recites "In a particular embodiment, the plurality of gases can include an oxygen-containing gas" [abstract]. Note that this disclosure recites the phrase "can include" indicating that the presence of an oxygen-containing gas (or air) is optional. Additionally, Frank explicitly  
15   discloses that the composition of gas within the sensing element may or may not comprise any oxygen-containing gas [0037]. As such, it is evident that Frank teaches embodiments of the disclosed detector wherein no oxygen-containing gas, moisture, or air is present.

          Additionally, as is evidenced by the Knoll reference cited by Applicant in the  
20   Response, one of ordinary skill in the art would have understood that the presence of air is the most typical cause of failure in proportional counters (Knoll - page 534). As such, one of ordinary skill in the art would have considered not including air in the detector gas of the neutron sensitive detector disclosed by Frank.

Finally, at page 14 of the Response Applicant opines that you can have no electronegative gases in neutron detectors, however the instantly pending claims include gas mixtures which include CO<sub>2</sub> (see instantly claims 5, 9, 10, 16, 17, and 24).

Of note with regards to CO<sub>2</sub> is that the molecule comprises two polar covalent C-O

5 bonds wherein the shared electrons are pulled towards the oxygen atom due to oxygen's high electronegative (i.e., 3.5) compared to that of carbon (i.e., 2.5). Thus, CO<sub>2</sub> is understood to be an electronegative gas. This begs the question, if the Applicant's assertion that neutron detectors cannot comprise electronegative gases, then is the claimed CO<sub>2</sub> gas containing neutron detector enabled? This point  
10 notwithstanding, given that Frank plainly teaches embodiments wherein there is no oxygen-containing gas, the Applicant's argument is not found persuasive.

On pages 16-18 of the Response, Applicant argues that Zhou does not overcome the deficiencies of Frank. However, the Examiner maintains that Frank is not deficient. Zhou is relied upon in the rejections primarily to show that boron 10 can be  
15 used in detectors, where Frank only uses boron. None of Applicant's discussion about Zhou relates to this teaching and therefore is not relevant to the rejection. Applicant is arguing the references individually, rather than arguing against the combination.

One point the Examiner must bring is that Applicant argues Zhou teaches away due to silence. This is a clear misstatement of the law, as mere silence or even  
20 disclosure of alternatives is not a teaching away; rather, for a teaching away a reference must criticize, discredit, or otherwise discourage following the path set out in the reference or the claimed invention. *In re Kahn*, 441 F.3d 977, 990 (Fed. Cir. 2006); *In re*

*Fulton*, 391 F.3d 1195, 1201 (Fed. Cir. 2004). Applicant does not point out any criticism or discouragement.

On page 18 of the Response the Applicant argues Lacy teaches away for various reasons. These arguments are not found persuasive as Lacy clearly is only relied upon to teach that the optimal thickness in boron 10 detectors is 1 mg/cm<sup>2</sup>. To the extent that Lacy teaches about different types of detectors elsewhere, such teachings are not relevant to Frank, the part of Lacy that is being relied upon, or the combination. Again, Lacy does not in any way discourage or discredit the material used for the combination.

### ***Conclusion***

#### **Claims 1-41 are rejected.**

Applicant is reminded of the continuing obligation under 37 CFR 1.178(b), to timely apprise the Office of any prior or concurrent proceedings in which Patent No. 10,197,686 is or was involved. These proceedings would include any trial before the Patent Trial and Appeal Board, interferences, reissues, reexaminations, supplemental examinations, and litigation.

Applicant is further reminded of the continuing obligation under 37 CFR 1.56, to timely apprise the Office of any information which is material to patentability of the claims under consideration in this reissue application.

This obligations rest with each individual associated with the filing and prosecution of this application for reissue. See also MPEP § 1404, 1442.01, and 1442.04.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE  
5 MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be  
10 calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to LEE E SANDERSON whose telephone number is  
15 (571)270-1079. The examiner can normally be reached M-F: 9:30AM to 7:00PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

20 If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Patricia Engle can be reached at 571-272-6600. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.



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